

Project Report: SnapClass AI Attendance System

1. Introduction

SnapClass is a modern, AI-powered attendance management system built to eliminate the traditional, time-consuming roll-call process. By leveraging Facial Recognition and machine learning, SnapClass instantly marks attendance for an entire classroom from a single uploaded photo.

2. Key Features

For Teachers:

- Class Management: Easily create new classes and generate unique shareable join codes, including automatic QR code generation.
- AI Face Attendance: Upload one or more photos of the classroom. The AI automatically scans, detects, and marks present students in seconds.
- Attendance Records: Keep track of historical attendance logs securely.

For Students:

- Face ID Login & Registration: Students securely log in and register their initial face profile using their webcam directly in the app.
- Auto-Enrollment: Seamlessly enroll in classes using a teacher's unique join link or code.
- Track Progress: View enrolled subjects and track personal attendance records across all classes.

3. Technical Architecture

The system is built on a robust, modern technology stack:

- Frontend & UI: Developed using Streamlit for a responsive, fast, and interactive web interface.
- Backend Database: Powered by Supabase (PostgreSQL), utilizing Row Level Security (RLS) to ensure data privacy and secure access controls.
- Artificial Intelligence (AI): Built using dlib and face_recognition_models for high-accuracy face detection and 128-dimensional facial embedding extraction. Uses scikit-learn Support Vector Classifier (SVC) for matching detected faces against the enrolled student database.
- Data Processing: Utilizes numpy for rapid mathematical array operations and threshold matching, along with Pillow for image handling.

4. System Workflow

- 1. Enrollment:** A student captures a selfie using the webcam. The system extracts a facial embedding and stores it securely in Supabase.
- 2. Class Creation:** A teacher creates a subject, generating a join code.
- 3. Student Joining:** Students use the code or scan the QR code to enroll in the specific subject.

4. Attendance Marking: The teacher uploads a classroom photo. The system detects all faces, extracts their embeddings, and runs them against the trained SVC model of enrolled students to mark attendance.

5. Conclusion

SnapClass successfully demonstrates the integration of advanced machine learning models into a user-friendly web application, providing an efficient, scalable, and secure solution for classroom attendance tracking.